

Relationship between the spectrometric values of deoxynucleic acid, ribonucleic acid and the PCR presence of a pathogen in single tick samples

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Abstract

Aim: Ticks are the vectors of many pathogens, which cause diseases with fatal consequences. Polymerase chain reaction (PCR) is being used to detect the presence of these different pathogens in ticks, but there is a need of isolated nucleic acid to conduct the molecular assays. In our previous research, we found that some ticks give huge yield of isolated nucleic acid during spectrometric measurements, therefore aim of this study is to find whether there is any relation between spectrometric values of DNA, RNA and presence of *Borrelia burgdorferi* as example pathogen in single tick samples.

Method: DNA and RNA were isolated with mini column method from single tick samples. They were run in real time as well as conventional PCR tests for the presence of *Borrelia burgdorferi*. The nucleic acid yields of isolated nucleic acid samples were measured with a spectrophotometer.

Results: It was found that there were 47 ticks positive for *Borrelia burgdorferi* and 40 were negative. Average isolated DNA and RNA quantity was higher in pathogen positive ticks than those of negative ticks. There was no correlation between the yield of nucleic acid and presence of pathogen in a single tick, but there was tendency that pathogen positive tick gave higher yield of DNA and RNA during the isolation.

Conclusions: This study shows some of *Borrelia burgdorferi* positive ticks give very high yield of DNA and RNA during the isolation. There is no correlation between presence of pathogen and nucleic acid in a single tick, but there is tendency that pathogen positive tick may have higher nucleic acid yield. Therefore, our recommendation is that laboratory should always measure the nucleic acid yield along with conducting the PCR tests.

Keywords: *Borrelia burgdorferi*, nucleic acid, single tick, polymerase chain reaction

Ticks are the vectors of a wide range of pathogens, which cause a number of fatal diseases in human beings. Therefore, it is essential to monitor such pathogens. Polymerase chain reaction (PCR) is one of the important methods being used widely for detection of various microbes today. (1,2,3,4) It makes possible to detect them in ticks. There are many publications about the detection of pathogens in ticks, but many research groups are using different methods to isolate nucleic acid from ticks, most commonly the pooled ticks. (5,6,7,8,9,10,11,12, 13,14)

Our group has developed a mechanical crushing method to isolate the DNA / RNA from a SINGLE tick. During the publication of this method, we found that there may be some co relation between the presence of high yield of DNA / RNA in a single tick with the presence of a pathogen in it. (1) In this work, we used *Borrelia burgdorferi* (B.b.) as an example. In the literature, there are rare reports of spectrometric values of a larger number of nucleic acid samples in the ticks as most of groups are using pooled samples to detect the presence of pathogens in ticks. (5,15) It is very important to find the presence of pathogens in a SINGLE tick to develop the accurate preventive and therapeutic

strategy in a particular area. We have published a simple and inexpensive method to isolate the nucleic acid from a SINGLE tick, which may open new opportunities. (1)

Therefore, in this research work, we decided to conduct further studies to find whether there is any correlation between the spectrometric values of isolated DNA / RNA and PCR presence of *Borrelia burgdorferi* in single tick samples.

Methods: The ticks were sent with a letter at room temperature with German postal service. DNA and RNA were isolated with mini column isolation kit. The mechanical crushing method was used. This is described in other publication fully. (1).

The real time and conventional PCR kits (Genekam, Germany) were used to detect the presence of *Borrelia burgdorferi* in the isolated nucleic acid samples. The machine used for real time PCR was ABI 7500 (Thermofischer) and the results were read at Ct values along with presence of the curves. Conventional PCR was conducted in thermocycler (Biometra, Germany) and the results were seen as band in the gel agarose stained with ethidium bromide. Positive and negative samples were used. (1)

The yield of isolated DNA and RNA from each single tick was measured in Nanodrop (Thermofischer, USA). The spectrometer was calibrated with elution buffer of the nucleic acid isolation kit. After that 3 values per sample (single tick) were measured and the average of these values was calculated. These average value of all B.b. positive and negative samples are shown in table 1 and 2.

Results: Total number of 47 B.b. positive single ticks were used for measuring the nucleic yield. The average of isolated DNA and RNA from these 47 ticks was 80.27 ng / μ l and 60.42 ng / μ l per single tick respectively. These average values show that yield of DNA per tick is more than that of RNA. The DNA yield per sample varies from minimum 0.2 ng / μ l to maximum 856.7 ng / μ l, hence there is a wide variation of isolated DNA yield between the single tick samples. Similarly, there was a broader variation range of RNA yield, where the minimum value varies from -0.15 to maximum value 591.3 ng / μ l.

Total number of 40 B.b. negative single ticks were used for measuring the nucleic acid outputs. It is found that average DNA yield each tick was 57.11 ng / μ l, where the average yield per tick for isolated RNA was 41.52 ng / μ l. These results show that DNA yield in B.b. negative samples was lower than that of RNA. These results have similar pattern as with those of B.b. positive samples. The DNA yield varies 0 to 646.5 ng / μ l. One sample, which was B.b negative, but it was positive for tick borne encephalitis virus. (Data not shown) This sample has DNA yield 2.0 ng / μ l and RNA yield was 1.3 ng / μ l. RNA yield varies between from minimum -0.3 to maximum 296.1 ng / μ l.

Three different groups depending upon the yield of isolated nucleic acid were created to find the percentage range among these groups. These were 0-20 ng / μ l, 20-200 ng / μ l and more than 200 ng / μ l. The number of samples with more than 200 ng / μ l DNA among positive tick were 5 / 47 total = 10.64% against those with more than 20 ng / μ l DNA up to 200 ng / μ l DNA yield were 20, which is 40.43%. Highest percentage was found in the group lower than 20 ng / μ l. (Fig 1) It was found that 68.18% negative ticks are under 20 ng / μ l. (Fig 2)

Number of ticks with higher DNA and RNA yield are more in B.b. positive ticks against the number of B.b. negative tick. This can be expressed as percentage and the probability, hence there is tendency that higher yield of DNA and RNA are in B.b. positive samples. There is no true co relation between the nucleic acid yields in B.b. positive and negative samples.

Discussion: In this research work, we analyzed total 87 single tick samples to find a relation between spectrometric values of isolated DNA, RNA and B.b. presence in single tick samples. In our previous publication, where we found that there is hardly any publication about studies for the total yield of DNA and RNA from one single tick. In this publication, we thought that the highly DNA yield from one single tick should be corelated with presence of the *Borrelia burgdorferi*, but our results of this research work do not support fully this idea, whereas the results are supporting that there is a

tendency that higher DNA and RNA yields from single tick may be an indication of presence of the pathogen in a tick, therefore it is essential that user should measure nucleic acid yield from each tick. At present, most of research groups are using pooled sample, but these groups can use our single tick isolation method as standard, so the results can be compared easily.

In the literature, there are a few reports, where the spectrophotometric values are measured from ticks. In one report, there are yield of isolated DNA from pooled tick, which was 2000 ng / μ l as this is very high yield from pooled ticks. Such high yield may lead to false results in PCR as very high concentration of nucleic acid can lead to failure of PCR.

The results of this publication are showing also that it is essential to measure the yield of isolated nucleic acid so that user can use optimal concentration of DNA / RNA during the PCR analysis because too high concentration may lead to questionable results.

One comparison between the mRNA vaccine against the coronavirus and concentration of B.b. positive tick DNA, a vaccine dose contains 50 μ g in 500 μ l of solution (100 ng / μ l), but a tick has up to 856 ng / μ l and many ticks have more than 200 ng / μ l DNA in this study. It shows that some ticks contain many times more nucleic acid than the amount being used as vaccine today. There is urgent need of more research about the role of such high dose inoculated in the persons from the tick bite and how such doses effect the clinical outcome. There are many laboratories, which are publishing the research about the presence of pathogens in ticks as well as patients suffering from these infections. These should measure the spectrometric yields of isolated nucleic acid of such ticks to contribute for the better scientific understanding.

In these studies, we studied only one pathogen (*Borrelia burgdorferi*), but this work shows that one has to conduct the studies about the presence of other pathogens in order to have better overview.

This publication shows the importance of spectrometric values of nucleic acid yields from the ticks and some co relation or tendency with the presence of pathogens in a tick. It is highly recommended for other research workers to measure such values and conduct further studies of molecular analysis to find such correlations. The result report of tick analysis should also contain the yield of isolated nucleic acid along with the molecular detection of the presence of pathogens.

Conclusion: In this research work, first time, we studied to establish a correlation between the spectrometric value of nucleic acid (DNA and RNA) of a single tick and presence of the pathogens. This indicates that there is a tendency of presence of pathogen in high nucleic acid yielding ticks, but the studies were unable to establish a real correlation between the higher nucleic acid yield and presence of *Borrelia burgdorferi* in single tick. It is highly recommended to measure the spectrometric values of isolated nucleic acid along with molecular detection of pathogens, hence both parameters should be the parts of tick analysis report. Such parameters may lead to develop better understanding the clinical outcome of tick-borne infections as well as control of these tick infections.

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Table 1 shows the spectrometric values of DNA and RNA yield of different pathogen positive single ticks.

Tick No.	DNA ng / μ l	RNA ng / μ l	Tick No.	DNA ng / μ l	RNA ng / μ l
1	856.7	591.3	25	19.1	15.6
2	556.8	303.3	26	19	44
3	292.8	312	27	17.5	9.7
4	245.3	68.5	28	17.2	12.7
5	212.5	170.9	29	15.5	13.5
6	194.3	189.4	30	13.8	12.4
7	188.2	155.5	31	13.3	6
8	184.3	152.1	32	12.6	9.4
9	110.6	89.1	33	10.7	8.6
10	97.4	76.9	34	9.5	7.2
11	97.2	94.8	35	5.1	4.7
12	83.3	62	36	5.1	5.6
13	72.6	59.9	37	4.9	4.3
14	66.5	52.1	38	4.4	6.3
15	62.4	50.6	39	4.1	3
16	51.6	30.2	40	3.7	1.2
17	41.7	37.7	41	3.6	4
18	41.1	25.2	42	2.1	2
19	29.5	16	43	2	2.2
20	26.7	23.1	44	1.7	2.1
21	26.1	20.8	45	1.2	1
22	24.7	20.2	46	0.2	-0.15
23	20.3	16.6	47	-16.2	34.3
24	20.2	12.1			

Table 2: shows the spectrometric values of DNA and RNA yields of different *Borrelia burgdorferi* negative single ticks.

Tick No.	DNA ng / μ l	RNA ng / μ l
1	1.2	1
2	17.2	12.7
3	8.15	5.3
4	3.2	2.3
5	54.2	57.9
6	83.2	37.4
7	50.3	77,4
8	58.3	108.5
9	283.2	149.9
10	79.8	59.6
11	6.3	16.3
12	7.1	9.2
13	646.5	367.1
14	1.1	0.7
15	7.5	5
16	13	12.1
17	5.7	4.4
18	2	1.3
19	157.5	68.6
20	65.4	50.8
21	8.8	7.9
22	0.7	0.6
23	2,5	0.8
24	5.4	3.7

Tick No.	DNA ng / μ l	RNA ng / μ l
25	4.4	3.5
26	2.3	2.5
27	3	2.9
28	3.6	3.9
29	33.8	20.7
30	5.1	5.1
31	7,2	4.5
32	355.8	296.1
33	340.4	271.1
34	20	14.8
35	16.2	11.9
36	4.8	3.3
37	0	-0,3
38	57.9	48
39	12.5	9.3
40	3,6	3.7
41	1	0.6
42	14.4	11.5
43	53.3	47.4
44	5.5	5.7

Fig. 1 shows the percentage of different DNA yields of *Borrelia burgdorferi* positive ticks.

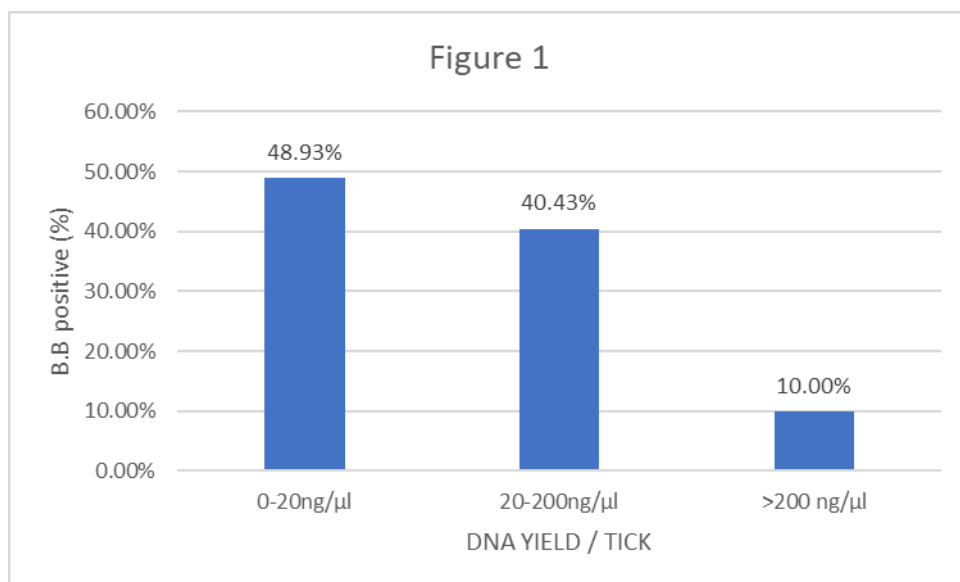


Fig 2: shows the percentage of different DNA yields of *Borrelia burgdorferi* negative ticks.

